

```
: # Analysis Starts
#Promotion Ready Employees by Department
df.groupby('department')['promotion_ready'].value_counts()
```

```
: department      promotion_ready
Human Resources   No                33
                  Yes                30
Research & Development Yes          527
                  No                434
Sales             Yes                370
                  No                76
Name: count, dtype: int64
```

```
: #Work Stability by Age Group
df.groupby('agegroup')['work_stability_score'].mean()
```

```
: agegroup
18-25    1.813008
26-35    5.521452
36-45    9.655983
46-55   16.654867
55+     16.382979
Name: work_stability_score, dtype: float64
```

```
6
7      -- 1 Total employees
8 •    SELECT COUNT(*) AS total_employees FROM hr_data;
9
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	total_employees			
▶	1470			

```
10      -- 2 Attrition count
11 •    SELECT attrition, COUNT(*)
12      FROM hr_data
13      GROUP BY attrition;
```

Result Grid		Filter Rows:	Export:
	attrition	COUNT(*)	
▶	Yes	237	
	No	1233	

```
15 -- 3 Department wise employees
16 • SELECT department, COUNT(*)
17 FROM hr_data
18 GROUP BY department;
```

Result Grid   Filter Rows: Export 

	department	COUNT(*)
▶	Research & Development	961
	Sales	446
	Human Resources	63

```
20 -- 4 Average salary by department
21 • SELECT department, AVG(monthlyincome)
22 FROM hr_data
23 GROUP BY department;
```

Result Grid   Filter Rows: Export: 

	department	AVG(monthlyincome)
▶	Research & Development	6281.2529
	Sales	6959.1726
	Human Resources	6654.5079

```

25      -- 5 Highest paying job roles
26  •    SELECT jobrole, AVG(monthlyincome) AS avg_salary
27      FROM hr_data
28      GROUP BY jobrole
29      ORDER BY avg_salary DESC;

```

Result Grid   Filter Rows: Export:  Wrap Cell Content:

	jobrole	avg_salary
▶	Manager	17181.6765
	Research Director	16033.5500
	Healthcare Representative	7528.7634
	Manufacturing Director	7295.1379
	Sales Executive	6924.2791
	Human Resources	4235.7500
	Research Scientist	3239.9726
	Laboratory Technician	3237.1699
	Sales Representative	2626.0000

```

31      -- 6 Attrition by department
32  •    SELECT department, attrition, COUNT(*)
33      FROM hr_data
34      GROUP BY department, attrition;
35

```

Result Grid   Filter Rows: Export:  Wrap Ce

	department	attrition	COUNT(*)
▶	Research & Development	Yes	133
	Sales	No	354
	Sales	Yes	92
	Research & Development	No	828
	Human Resources	Yes	12
	Human Resources	No	51

```

36      -- 7 Overtime impact on attrition
37 •    SELECT overtime, attrition, COUNT(*)
38      FROM hr_data
39      GROUP BY overtime, attrition;

```

Result Grid   Filter Rows: Export:  Wrap Cell C

	overtime	attrition	COUNT(*)
▶	No	Yes	110
	No	No	944
	Yes	Yes	127
	Yes	No	289

```

41      -- 8 Gender distribution
42 •    SELECT gender, COUNT(*)
43      FROM hr_data
44      GROUP BY gender;

```

Result Grid   Filter Rows:

	gender	COUNT(*)
▶	Male	882
	Female	588

```

46      -- 9 Average experience by department
47 •    SELECT department, AVG(totalworkingyears)
48      FROM hr_data
49      GROUP BY department;

```

Result Grid   Filter Rows: Export:  Wrap Cell C

	department	AVG(totalworkingyears)
▶	Research & Development	11.3424
	Sales	11.1054
	Human Resources	11.5556

```

51      -- 10 Salary slab distribution
52  •    SELECT salaryslab, COUNT(*)
53      FROM hr_data
54      GROUP BY salaryslab;

```

Result Grid



Filter Rows:

	salaryslab	COUNT(*)
▶	Upto 5k	749
	5k-10k	440
	10k-15k	148
	15k+	133

```

56      -- 11 Job satisfaction by job role
57  •    SELECT jobrole, AVG(jobsatisfaction)
58      FROM hr_data
59      GROUP BY jobrole;

```

Result Grid



Filter Rows:

Export:



Wrap Cel

	jobrole	AVG(jobsatisfaction)
▶	Laboratory Technician	2.6911
	Sales Representative	2.7349
	Research Scientist	2.7740
	Human Resources	2.5577
	Manufacturing Director	2.6828
	Sales Executive	2.7546
	Healthcare Representative	2.7863
	Research Director	2.7000
	Manager	2.7059

```

61      -- 12 Employees with highest salary hike
62 •    SELECT empid, monthlyincome, percentsalaryhike
63      FROM hr_data
64      ORDER BY percentsalaryhike DESC;

```

Result Grid   Filter Rows: Export:  Wrap Cell Content:

	empid	monthlyincome	percentsalaryhike
▶	RM854	2552	25
	RM893	1859	25
	RM1308	3591	25
	RM121	2613	25
	RM435	10648	25
	RM1076	11244	25
	RM1199	5368	25
	RM1088	2308	25
	RM677	4014	25
	RM1433	13744	25
	RM252	10938	25
	RM1373	5151	25

hr_data 12 ×

Output

```

66      -- 13 Most experienced employees
67 •    SELECT empid, totalworkingyears
68      FROM hr_data
69      ORDER BY totalworkingyears DESC;

```

Result Grid   Filter Rows: Exp

	empid	totalworkingyears
▶	RM127	40
	RM596	40
	RM099	38
	RM271	37
	RM446	37
	RM086	37
	RM1302	37
	RM895	36
	RM1185	36
	RM188	36
	RM1117	36
	RM402	36

```

71      -- 14 Attrition by marital status
72 •    SELECT maritalstatus, attrition, COUNT(*)
73      FROM hr_data
74      GROUP BY maritalstatus, attrition;

```

Result Grid   Filter Rows: Export: 

	maritalstatus	attrition	COUNT(*)
▶	Single	Yes	120
	Single	No	350
	Divorced	No	294
	Married	No	589
	Married	Yes	84
	Divorced	Yes	33

```
76 -- 15 Education field wise salary
77 • SELECT educationfield, AVG(monthlyincome)
78 FROM hr_data
79 GROUP BY educationfield;
```

Result Grid			Filter Rows:	Export:	Wrap C
	educationfield	AVG(monthlyincome)			
▶	Life Sciences	6463.2888			
	Medical	6510.0366			
	Marketing	7348.5849			
	Technical Degree	5758.3030			
	Other	6071.5488			
	Human Resources	7241.1481			